

Turn Your Old Inverter Into An Emergency Power System

An inverter turned into emergency power system, which turns on when the mains supply fails, and more importantly doesn't turn on when the main supply is available

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Can you give me a setup (emergency lighting system) where few of my lights will turn on when the supply from mains is unavailable, and more importantly, will not turn on when the supply from mains is available? That's the question a reader asked us, who wanted to set up an emergency lighting system in his factory.

Our first reaction was "why doesn't he use an inverter?" But then we realised that a normal inverter would mean that the emergency light would be turned on when the power from main is there too—something the reader did not require. Quickly, we dished out an old inverter and made a few changes. We hooked our customised inverter to the wire powering his emergency light. It took us less than two hours to reconfigure the inverter, and we got a happier reader. Now we share with you the changes made by us, in the hope to get even more happy readers.

Construction

There are several ways to make the automatic on/off switching circuit for the inverter, but to make it simple and universal is slightly challenging. To make a universal circuit, a transformer, a regulator, a relay and a few discrete components are required. Fig. 1 shows the circuit of the home-made auto-on/off switch. The circuit assembled on a general-purpose PCB is shown in Fig. 2.

The circuit was tested on an APC 800VA inverter, which has 230V AC input and output sockets (refer Fig. 3).

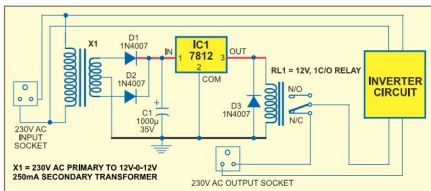


Fig. 1: Auto on/off circuit for inverter



Fig. 2: Circuit assembled on a general-purpose PCB



Fig. 3: Rear panel of APC 800VA inverter showing input cord and output socket